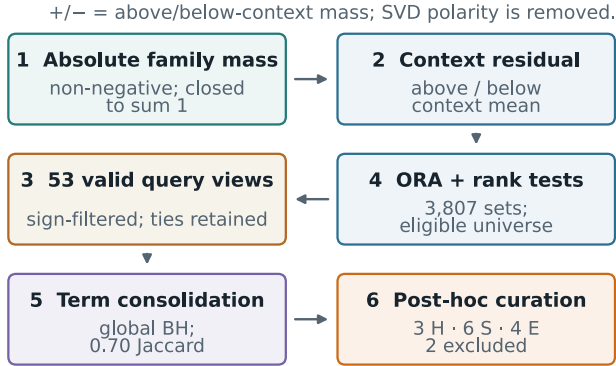


a Operational definition and audit workflow



b Nine manually curated annotations

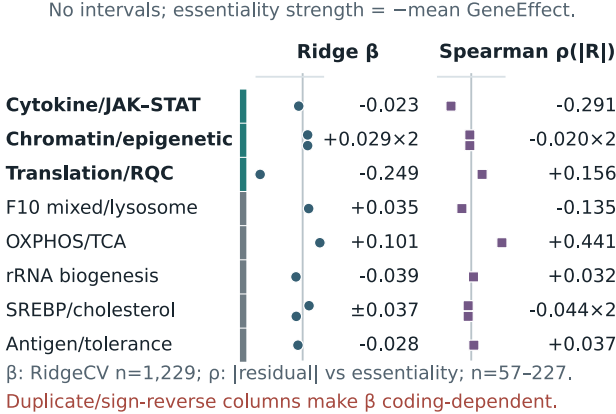
Corrected $-\log_{10}(q)$; columns are dependent summaries.

		annot.†		fitness‡		study
		ORA	rank	lpl	lpl	max %
F14 lower	Cytokine/JAK-STAT	54.7	16.0	0.023	0.291	100%
F3/15 HCabs	Chromatin/epigen.	14.5	NR	0.029	0.020	88%
F12 low/abs	Translation/RQC	9.9	21.1	0.249	0.156	100%
F10 HC/a/u	Mixed chr.-cytokine	39.0	21.5	0.035	0.135	100%
F7 upper	OXPHOS / TCA	8.9	11.1	0.101	0.441	100%
F10 lower	Lysosome / ECM	5.3	4.3	0.035	0.135	100%
F11 low/abs	rRNA biogenesis	4.3	3.3	0.039	0.032	100%
F5 low/F6 up	SREBP/cholesterol	6.1	5.1	0.037	0.044	100%
F1 abs/low	Antigen/tolerance	3.1	3.6	0.028	0.037	64%

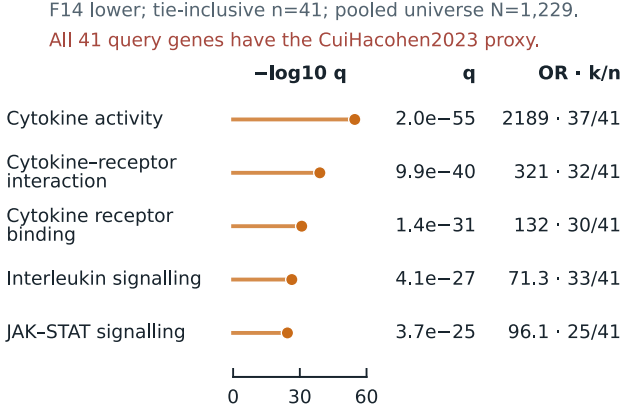
† ORA/rank share inputs; ‡ β/p share GeneEffect.

Shade per column; study = maximum study-proxy share.

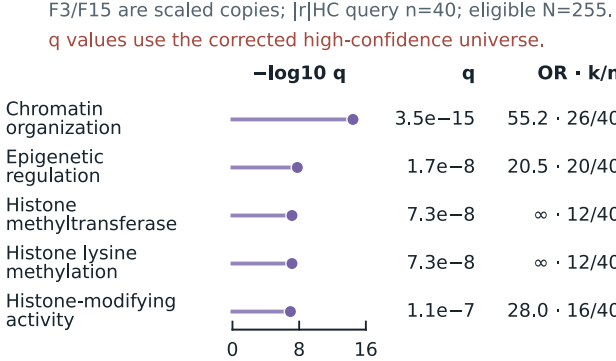
c Shared-endpoint fitness summaries



d Study-conditioned cytokine coherence



e Deduplicated chromatin coherence



f Why these are not general biological axes

Frozen outputs lack study-held-out reconstruction.

F14 lower

41 tied genes; 100% CuiHacohen2023 study-conditioned cytokine coherence

F3/F15 HCabs

F15 = 0.193548 $\times$ F3; 88% LaraAstasioHuntly2023 one candidate, not two replications

F12 abs/lower

abs 40/40; lower 30/30; 100% ReplogleWeissman2022 K562-conditioned translation/RQC

Fitness β

16 closed features; rank 12; F6 = -F5; F4/F13 = 0 β is coding/penalty-dependent

Needed: leave-study-out rebuilds · matched nulls · contrast-basis bootstrap